Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/11/2011
PI Name:	Yokota, Hiroki Ph.D.		
Project Title:	Integrated Stress Response in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Organization Name:	Indiana University Purdue University Indianapolis		
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PI Web Page:			
City:	Indianapolis	**State:**	IN
Zip Code:	46202	**Congressional District:**	7
Comments:	NOTE: PI moved back to Indiana University from Rensselaer Polytechnic Institute in January 2011 [Ed., 01/26/2011]. PI previously at Indiana University until August 2010 [Ed. 09/2010]		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	09/14/2010	**End Date:**	03/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: Administrative PI is Eric Ledet, Rensselaer Polytechnic Institute (RPI). PI Yokota moved back to Indiana University Purdue University Indianapolis from RPI in January 2011. PI originally at Indiana University Purdue University Indianapolis and moved to RPI in August 2010. [Ed., 04/05/2011]		
COI Name (Institution):	Zhang, Ping (Indiana University) Ledet, Eric (ADMINISTRATIVE PI: Rensselaer Polytechnic Institute) Zhao, Liming M.D. (Indiana University)		
Grant/Contract No.:	NNX10AU51G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note: 4/05/2011] PI Yokota moved to Rensselaer Polytechnic Institute (RPI) in August 2010 and this grant (NNX10AU51G) was established at that time. PI has since returned to Indiana University Purdue University Indianapolis and the research is being conducted through a sub-contract from RPI (Administrative PI is Eric Ledet) to Indiana University Purdue University Indianapolis (PI is Hiroki Yokota). We address a set of questions: Does unloading evoke translational reprogramming through ISR-linked phosphorylation of eukaryotic initiation factor 2a (eIF2alpha)? If so, what mechanisms are responsible for selecting an ISR-linked pro-survival or apoptotic pathway? We hypothesize: • Transient unloading induces an ISR-linked pro-survival program by activation of eIF2alpha phosphorylation and reduction in translational efficiency. • Continuous unloading induces oxidative stress and apoptosis through accumulation of stress to the ER, but this chronic stress effect can be alleviated through a sustained phosphorylation level of eIF2alpha To test the above hypothesis and examine a potential role of ISR-driven signaling under unloading, we propose the following three specific aims using a mouse hindlimb suspension model: • Aim 1: Evaluate phosphorylation of eIF2alpha and its kinases (e.g., PERK and GCN2) as well as translational efficiency in response to transient unloading • Aim 2: Evaluate gene regulatory mechanisms linked to oxidative stress and stress to the ER in response to continuous unloading • Aim 3: Examine the effects of administration of salubrinal (inhibitor of eIF2alpha phosphatase) on pro-survival signaling and bone metabolism The project will employ real-time PCR, Western blot analysis, enzymatic assays, and bone histology with pharmacological agents, and eIF2a-linked knockout mice. Research schedule: The original timetable is unchanged. The first year will be focused on Aim1, in which the role of eIF2alpha phosphorylation, PERK/GCN2 kinases, and gene regulation in unloading will be investigated. In the second and the third years, Aims 2 and 3 will be conducted, respectively.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The result in the first year supports the hypothesis that unloading is a novel form of stress sources that induce integrated stress responses (ISR). This finding allows us to consider a novel therapeutic treatment that mitigates ISR under microgravity. The finding also contributes to our fundamental understanding about ISR on ground.
Task Progress:	As planned for Aim 1 in the first year, the ongoing study employed C57BL/6 wildtype mice and GCN2 knockout mice (8240, Jackson Lab) (~ 14 weeks of age). In mechanical loading, a piezoelectric loader was employed to apply loads to the left ankle. The loading condition was 0.5 N at 5 Hz for 5 min. Mice were sacrificed 3 h after loading. In unloading, mice were treated with hindlimb suspension for 24 h. To examine the phosphorylation level of eIF2a in response to mechanical loading, we employed ankle loading and harvested tibiae at 3 h after the loading bout. Compared to the contralateral control, we observed that the loaded tibia presented a reduced level of eIF2a-p in both C57BL/6 (wildtype) and GCN2 knockout mice. To evaluate the effects of unloading on the modulation of eIF2a phosphorylation, we harvested tibiae from C57BL/6 mice at 24 h after tail suspension and conducted Western blot analysis. The results show that unloading upregulated the phosphorylated level of eIF2a. Unloading is a cause of disuse osteoporosis to bedridden patients as well as astronauts. The ongoing study demonstrates that unloading elevated the phosphorylated level of eIF2a, while mechanical loading downregulated it. Since eIF2a phosphorylation is induced by various stress sources that induce an integrated stress response (ISR), the result suggests that mechanical loading potentially suppresses ISR-linked stress. The phosphorylation of eIF2 alpha can be triggered by nutrient deprivation by GCN2. The result with GCN2 knockout mice indicate that the observed load-driven suppression of ISR is not mediated by GCN2.
Bibliography Type:	Description: (Last Updated: 04/09/2013)
Abstracts for Journals and Proceedings	Zhang P, Yokota H. "Loading and unloading-driven regulation of phosphorylation of eIF2a" Presented at the 57th Annual Meeting of the Orthopaedic Research Society, Long Beach, CA, January 13-16, 2011. Transactions of the 57th Annual Meeting of the Orthopaedic Research Society, 2011 Feb;57. , Feb-2011
Articles in Peer-reviewed Journals	Zhang P, Jiang C, Ledet E, Yokota H. "Loading- and unloading-driven regulation of phosphorylation of eIF2a." Biological Sciences in Space. 2011 Apr;25(1):3-6. https://www.jstage.jst.go.jp/article/bss/25/1/25_3/_pdf , Apr-2011